

Prompting-based fake news detection framework for Bengali

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Abstract

Due to the lack of reliable detecting techniques and annotation resources, the quick spread of false information in Bengali digital media has become a serious problem. In this study, a hybrid evaluation of traditional machine learning methods and transformer-based architectures was conducted for Bengali fake news detection. Multiple prompting strategies, such as zero-shot, few-shot, and chain-of-thought prompting, were used to evaluate models on a balanced dataset of Bengali news items. The experimental results demonstrated that traditional classifiers achieved moderate accuracy but struggled with recall and F1-scores in fake news classification. Transformer-based approaches yielded more consistent improvements, with the T5 model showing notable superiority. Specifically, T5 with chain-of-thought prompting outperformed other transformer models and conventional baselines in real news detection, with an accuracy of 92.5%, precision of 100%, recall of 85%, and F1-score of 91.9%. Large language model methods prompted by questions offer a cost-effective method for the detection of misinformation based on less data. They can be effectively applied to Bengali, a low-resource language, and the same strategy could be applied to other multilingual environments.

Keywords: Fake News Detection, Large Language Models, Low-Resource Languages, Prompt Engineering, Bengali NLP

1 Introduction

The fast growth of digital news and social media has multiplied the dissemination of false information, with serious impacts on social trust, public safety, and democratic deliberation. In Bengali society, misinformation has a significant social impact, affecting elections, disseminating communal unease, and derailing public awareness campaigns [1]. Even though Bengali is spoken by more than 230 million individuals across the globe, fake news detection is a significant issue because of the lack of annotated resources, linguistic equipment, and effective detection models. For detecting misinformation, standard machine learning and deep learning methods have been employed [2, 3]. Yet they predominantly depend on manually crafted features and large annotated corpora, which are expensive to obtain for low-resource languages such as Bengali. Additionally, these models tend not to generalize across domains because of linguistic variation and cultural difference [4, 5]. Low-resource languages now enjoy a higher number of alternatives due to recent progress in large language models (LLMs) [6], particularly transformer-based models such as BERT [7] and GPT variants [8]. Prompt engineering, where well-designed prompts condition model predictions, has been successful in low-resource environments by minimizing the necessity of expensive fine-tuning [9, 10]. More recent works demonstrate that prompting-based approaches can significantly enhance cross-lingual performance and improve zero-shot and few-shot classification for misinformation detection in underrepresented languages, including Bengali [11–13].

This study proposes a prompting-based paradigm for detecting fake news in Bengali. Transformer-based models are used to analyze various prompt design tactics, such as zero-shot, few-shot, and chain-of-thought prompting. A balanced Bengali fake news dataset is curated, and comparative experiments with traditional and transformer-based baselines are conducted. The results show that prompting with LLMs, particularly T5 with chain-of-thought reasoning, substantially improves detection performance, highlighting prompting as a scalable and data-efficient solution for misinformation detection in Bengali and other low-resource languages.

2 Early Work

Early research on automatic fake-news detection often combined linguistic content with social context features. The Paper [2] pioneered credibility assessment in Twitter by analyzing both textual content and propagation patterns. The Paper [14] demonstrated that Facebook “like” patterns alone could effectively distinguish hoaxes from genuine posts. The Paper [15] introduced the CSI model (Capture–Score–Integrate), a hybrid deep-learning framework that fuses content features, user behaviors, and source characteristics for rumor classification. In the same year, the paper [3] released the LIAR dataset—consisting of over 12,000 manually labeled political statements—providing a widely adopted benchmark for statement-level fact-checking research. Surveying these early methodologies, the paper [16] offered a comprehensive overview of content-based, propagation-based, and hybrid detection techniques, while the paper [17] synthesized emerging trends and evaluation gaps. Focusing specifically on

Bangla/Bengali, the paper [4] introduced BanFakeNews, a curated dataset of approximately 50,000 Bangla news articles labeled as real or fake, and established baseline detection models—marking one of the first large-scale efforts targeting Bengali fake-news detection. More recently, the rise of prompt-based methods and large language models has begun to influence Bengali misinformation detection research. Zero-shot fake news detection for Bengali was explored through cross-lingual prompts and multilingual pretrained models [11]. Few-shot prompting for hate speech and fake news detection in Bengali was shown to improve performance over zero-shot methods [12]. Chain-of-thought prompting has also been applied for reasoning-based tasks in Bengali, including fake-news justification, demonstrating the potential of interpretable outputs [13]. These studies indicate a transition from traditional supervised approaches to prompt-driven architectures in Bengali NLP, though systematic benchmarking and broader domain evaluations remain open challenges.

Table 1: Summary of selected early and recent works related to fake-news detection.

Core Idea	Main Findings	Limitations
Prompt-based few-shot paradigm combining textual prompts with social-graph alignment for fake-news detection [18]	Improves few-shot detection by aligning prompt templates with social topology.	Tested mainly on English; adaptation to Bengali unexplored; depends on social-graph data often unavailable.
A zero shot, context-aware prompt engineering method that leverages GPT as a classification model for detecting fake news across multiple domains [19]	Careful prompt design enables effective zero-shot classifiers across domains.	Highly sensitive to wording; weaker in domain-specific tasks; not validated on Bengali.
Task-specific prompt-tuning with contextual augmentation for early/few-shot detection [20]	Prompt-tuning yields strong few-shot performance and leverages contextual features.	Focused on high-resource languages; Bengali use faces tokenization and vocabulary challenges.
Prompt templates and fine-tuning applied to Bangla PLMs for sequence classification [21]	Improves Bangla NLP tasks such as sentiment and categorization, showing potential for fake-news detection.	Not directly tested on fake-news; depends on availability of Bangla PLMs and careful prompt engineering.
BanFakeNews dataset (~50K Bangla news articles) with baseline classifiers [4]	Provides the largest early Bangla fake-news dataset and strong baselines.	Dataset has balance and topical limitations; predates prompt-based methods.
mBERT with feature augmentation for multilingual COVID fake-tweet detection [22]	Shows reasonable Bengali performance and demonstrates zero-shot transfer potential.	Domain limited to COVID tweets; not prompt-based; restricted topical scope.

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Table 1 – *Continued from previous page*

Core Idea	Main Findings	Limitations
Deep learning models (Bi-GRU, LSTM, CNN, BERT) applied to Bangla fake-news datasets [5]	Achieves strong accuracy on balanced datasets using supervised methods.	Relies on full supervision; not designed for few-shot or prompt-based detection.

3 Designing Effective Prompts for Fake News Detection

Prompt engineering enables large language models (LLMs) to effectively perform tasks in low-resource languages such as Bengali without costly fine-tuning. Employing approaches such as zero-shot, few-shot, and chain-of-thought (CoT) methods, input prompts may be strategically crafted to direct LLMs to perform tasks such as sentiment classification, misinformation detection, and hate speech detection.

3.1 Zero-Shot Prompting

A model can do a task it has not been well-trained for using only the commands in the prompt with zero-shot prompting. This approach is particularly worth it for Bengali, as there are limited annotated datasets available. It was previously demonstrated that GPT-based models can be made to do classification or translation tasks in Bengali using well-crafted prompts and without further training [11]. Its success is based on the pretrained knowledge of the model and the fact that it has cross-lingual semantic coverage.

3.2 Few-Shot Prompting

Few-shot prompting provides the model with limited annotated examples to indicate the task inside the prompt. Even a small number of annotated examples can greatly enhance the accuracy of models in low-resource environments. For example, [12] demonstrated that few-shot prompting with 3–5 Bengali examples enhanced the performance of hate speech detection tasks compared to zero-shot prompting. By exposing the model to minimal examples, this method helps it capture task-specific input–output patterns and generalize more effectively.

3.3 Chain-of-Thought Prompting

Chain-of-thought prompting structures the input so that the model is encouraged to reason through intermediate steps before generating the final answer [23]. While most research has focused on English, initial work in Bengali has begun to emerge. In particular, [13] explored CoT prompting for tasks such as multi-hop question answering and fake news justification in Bengali. Their findings suggested that including intermediate reasoning improves the explainability and coherence of outputs, though challenges remain in ensuring grammatical fluency and contextual grounding.

Overall, prompt-based strategies provide practical and efficient solutions for NLP tasks in Bengali, where annotated resources are limited. Zero-shot and few-shot

prompting reduce the dependency on large training datasets, while CoT prompting enhances reasoning capabilities and explainability. Continued research is required to refine prompt design and address persistent challenges such as tokenization, morphological richness, and contextual grounding in Bengali.

4 Dataset Description

The dataset consists of 1,000 Bangla news articles organized into four columns: category, headline, content, and label. The category column specifies the type of news, such as National, Crime, Sports, or Miscellaneous. The headline column contains short Bangla news titles, while the content column provides the full text of the news article, offering detailed context. Given that the label column is binary and numeric (0.0 or 1.0), it most likely denotes a categorization result, such as whether the news is truthful or false. Crucially, the dataset is complete because it contains no missing values. This dataset is generally well-structured and balanced, which makes it appropriate for problems involving natural language processing, such as identifying false news in Bangali.

Table 2: Sample records from the Balanced Bangla Fake News Dataset

Category	Headline	Content	Label
National	ঘূর্ণিঝড় ফণীর আঘাতে কলকাতা শহর টিকবে তো?	মানুষের স্পর্ধা দেখে বেশ অবাকই লাগে। সুপার সাইক...	0.0
Miscellaneous	একজন স্বামী যেভাবে চিড়িয়াখানার গরিলাকে জন্ম করলেন	স্বামী স্ত্রী গেছে চিড়িয়াখানায়। এক গরিলার খাঁচা...	0.0
Crime	দুই সাবেক সেনা কর্মকর্তার ফাঁসি, তিন জনের কারাদণ্ড	এনএসআইয়ের সাবেক মহাপরিচালক ব্রিগেডিয়ার জেনারেল ...	1.0

A few sample records from the Balanced Bangla Fake News Dataset, which is intended to research and identify fake news in the Bengali language, are shown in Table 2. A news item is represented by each row in the table, while the columns list the news’s salient features. The news topic or domain, such as National, Crime, or Miscellaneous, is indicated in the Category column. The title column in Bengali displays the news title, which gives a summary of the content. A brief overview or summary of the news piece, also in Bengali, is given in the Content column. Lastly, whether the news is fake news or is indicated in the Label column, where 1.0 denotes fraudulent news and 0.0 denotes real news. The chart, for instance, reveals that a news item regarding two former army commanders under the "Crime" category was given a label of 1.0, suggesting that it is a fake, whilst other items within the National and Miscellaneous categories were given a label of 0.0, indicating that they are authentic. Overall, the table provides an overview of the dataset’s structure, highlighting the range of news categories and the labeling for tasks that require the identification of misleading data.

5 Methodology

To address the issue of Bengali false news identification, we employed a hybrid strategy that combines state-of-the-art transformer-based architectures with conventional machine learning methods. This dual design has two goals: (i) to create interpretable supervised baselines based on feature-engineering, and (ii) to use pretrained language models’ contextual capability to improve performance in a low-resource setting. Transformer-based models and conventional classifiers make up the two complementing halves of the methodology.

5.1 Traditional Machine Learning Models

Classical algorithms were implemented to benchmark the performance of transformer-based systems. These models rely on handcrafted text representations such as TF-IDF vectors and bag-of-words features, which serve as the input for classification into real or fake news categories.

5.1.1 Logistic Regression

A popular linear model for binary classification is logistic regression [24]. It predicts class membership probabilities via a sigmoid activation. Its appeal lies in interpretability and efficiency, making it an effective baseline despite its limited capacity to capture non-linear relationships.

5.1.2 Decision Tree

The Decision Tree classifier recursively partitions the feature space to form a tree-like decision structure [25]. In this case, each internal node represents some decision rule, and leaves correspond to classification outputs. While highly interpretable, this model is prone to overfitting, especially when applied to high-dimensional sparse text data.

5.1.3 Random Forest

Random Forest produces a tree ensemble trained on bootstrapped samples using randomized feature subsets to address the overfitting problem of individual decision trees [26]. Predictions are aggregated through majority voting, resulting in improved generalization and stability.

5.1.4 Gradient Boosting

Gradient Boosting constructs an additive model where weak learners, typically shallow trees, are trained sequentially [27]. Each new learner focuses on correcting the errors of its predecessor. This iterative refinement allows the model to capture complex dependencies and often outperforms bagging-based ensembles like Random Forest in predictive accuracy.

5.2 Transformer-Based Architectures

While traditional models provide baseline comparisons, transformer-based models represent the core of this study. These architectures encode text into contextualized embeddings that account for semantic and syntactic dependencies, enabling superior handling of Bengali’s morphological complexity. Pretrained multilingual or monolingual transformers were adapted for classification via prompting strategies.

5.2.1 BERT

To learn bidirectional context, Bidirectional Encoder Representations from Transformers (BERT) uses a masked language modeling strategy [7]. Improved iterations have continuously improved text categorization task performance. In Bengali fake news detection, BERT supports nuanced word-level disambiguation within broader sentence contexts.

5.2.2 RoBERTa

By training on bigger datasets, eliminating the Next Sentence Prediction assignment, and fine-tuning hyperparameters, RoBERTa improves on BERT’s architecture [28]. These enhancements yield richer embeddings and higher downstream accuracy, making it particularly effective for cross-domain classification.

5.2.3 T5

The Text-to-Text Transfer Transformer (T5) treats all NLP problems under a text-to-text paradigm, mapping inputs (news content) directly to outputs (labels such as “real” or “fake”) [29]. This unified design, coupled with large-scale pretraining, allows the model to adapt flexibly across diverse NLP tasks. In our framework, T5 was tested with zero-shot, few-shot, and chain-of-thought prompting to assess its adaptability under minimal task-specific supervision.

5.3 Prompt Engineering for Bengali Fake News Detection

Given the resource limitations of Bengali, prompt engineering was central to optimizing transformer performance. Custom prompts were designed in Bengali, reflecting natural sentence structures, frequent code-mixing patterns, and domain-relevant terminology.

Three prompting paradigms were employed:

- **Zero-Shot Prompting:** Only task instructions in Bengali were given to the model; no labeled examples were included. This configuration assesses pretrained transformers’ cross-lingual generalization [11].
- **Few-Shot Prompting:** A handful (3–5) of annotated examples were embedded within prompts to guide model predictions, offering minimal but effective supervision [12].
- **Chain-of-Thought Prompting:** In this instance, before generating a final categorization, the models were directed to provide intermediate reasoning stages. By

revealing the line of thinking, this increased accessibility in addition to classification accuracy [13].

By integrating these paradigms, the methodology achieves a balance between generalization (zero-shot), minimal supervision (few-shot), and reasoning-driven interpretability (CoT), creating a robust framework for detecting misinformation in Bengali digital media.

6 Experimental Setup

All experiments were run on high-performance servers equipped with NVIDIA Tesla V100 (32 GB) and A100 (40 GB) GPUs, supported by dual Intel Xeon processors and 256 GB system memory. Model development and training were carried out using PyTorch along with the Hugging Face Transformers framework. The AdamW optimizer was applied, with learning rates tuned between $2e-5$ and $5e-5$, and batch sizes selected in the range of 16–32. To prevent overfitting, each model was trained for three to five epochs before being stopped early depending on validation loss. The training objective was cross-entropy loss, while the evaluation metrics were accuracy, precision, recall, and F1-score. Data was divided into 70% training, 15% validation, and 15% testing using a 5-fold cross-validation. Normal machine learning algorithms finished training in minutes, in contrast to transformer models like BERT and RoBERTa, which took 20 to 25 minutes per epoch. With an average of 35 to 40 minutes per epoch on the A100 GPU, the T5 model proved more expensive. The impacts of different prompting techniques on Bengali fake news categorization using T5 were explored as well in this study.

7 Result

The proposed technique describes how transformer-based models, like BERT, ROBERTA, and T5, can quickly determine which news is authentic or fake. To help three transformer-based models complete tasks, we employ three prompting strategies: first zero-shot, then few-shot, and then chain of thinking.

7.1 bengali Prompt for BERT Model

Zero-shot prompt for Bert Model:

A prompting approach that uses zero shots was created for the BERT model. The Prompt is:

এই সংবাদটি [MASK]। "দুই সাবেক সেনা কর্মকর্তার ফাঁসি, তিন জনের কারাদণ্ড"

Few-shot prompt for Bert Model:

The BERT approach involved the development of a few-shot prompting strategy.

The Prompt is:

Headline: "দুই সাবেক সেনা কর্মকর্তার ফাঁসি, তিন জনের কারাদণ্ড" Content: "এনএসআইয়ের সাবেক

মহাপরিচালক ব্রিগেডিয়ার জেনারেল ..."

Label: [MASK]

Chain-of-Thought prompt for Bert Model:

A prompting approach that uses chain-of-thought was created for the RoBerta model.

The Prompt is:

প্রথমে, সংবাদটির বিষয়বস্তু বিশ্লেষণ করুন।

- সংবাদটি কি বাস্তবসম্মত?
- এর উৎস কি বিশ্বাসযোগ্য?
- এতে কোনো অতিরঞ্জিত বা অসঙ্গত তথ্য আছে কি?

পরে, প্রতিটি প্রশ্নের উত্তর দিয়ে সিদ্ধান্তে আসুন।

শেষে, নিচের ফাঁকা স্থানটি পূরণ করুন:

এই সংবাদটি ফেক/রিয়েল।

7.2 bengali Prompt for RoBerta Model

Zero-shot prompt for RoBerta Model:

A prompting approach, zero-shot was created for the RoBerta model.

The Prompt is:

সংবাদ: "দুই সাবেক সেনা কর্মকর্তার ফাঁসি, তিন জনের কারাদণ্ড।"

এই সংবাদটি সত্য নাকি মিথ্যা? উত্তর দিন: <mask>

Few-shot prompt for RoBerta Model:

A prompting approach, Few-shot was created for the RoBerta model.

The Prompt is:

সংবাদ: "দুই সাবেক সেনা কর্মকর্তার ফাঁসি, তিন জনের কারাদণ্ড।" এই সংবাদটি <mask>:

Chain-of-Thought prompt for RoBerta Model:

A prompting approach, chain-of-thought was created for the RoBerta model.

The Prompt is:

সংবাদ: "দুই সাবেক সেনা কর্মকর্তার ফাঁসি, তিন জনের কারাদণ্ড।"

প্রথমে সংবাদটির বিষয়বস্তু বিশ্লেষণ করুন:

- সংবাদটি কি বাস্তবসম্মত?
- এর উৎস কি বিশ্বাসযোগ্য?
- এতে কোনো অসঙ্গত বা অতিরঞ্জিত তথ্য আছে কি?

শেষে, সিদ্ধান্ত: এই সংবাদটি <mask>

7.3 bengali Prompt for T5 Model

Zero-shot prompt for T5 Model:

A prompting approach, zero-shot was created for the T5 model.

The Prompt is:

সংবাদ: "দুই সাবেক সেনা কর্মকর্তার ফাঁসি, তিন জনের কারাদণ্ড।"

নির্ণয় করুন: এই সংবাদটি কি মিথ্যা নাকি সত্য? উত্তর দিন শুধুমাত্র "ফেক" অথবা "রিয়েল" হিসাবে।

Few-shot prompt for T5 Model:

A prompting approach, Few-shot was created for the T5 model.

The Prompt is:

সংবাদ: "দুই সাবেক সেনা কর্মকর্তার ফাঁসি, তিন জনের কারাদণ্ড।"

বিশ্লেষণ: এটি অযৌক্তিক এবং অবাস্তব।

ফলাফল:

Chain-of-Thought prompt for T5 Model:

A prompting approach, chain-of-thought was created for the T5 model.

The Prompt is:

সংবাদ: "দুই সাবেক সেনা কর্মকর্তার ফাঁসি, তিন জনের কারাদণ্ড।"

প্রথমে, সংবাদটি বিশ্লেষণ করুন:

- এটি কি বাস্তবসম্মত বা বিশ্বাসযোগ্য উৎস থেকে এসেছে?
- এতে কোনো অতিরঞ্জিত বা অসঙ্গত তথ্য আছে কি?

শেষে, সিদ্ধান্ত দিন: এই সংবাদটি ফেক বা রিয়েল.

7.4 Result Analysis

While ensemble approaches (Random Forest and Gradient Boosting) and classic classifiers like Logistic Regression struggle with recall and F1-score, transformer-based architectures beat them in terms of accuracy for Bengali false news identification (shown in Table 3). On the other hand, transformer models show erratic outcomes. BERT and RoBERTa perform moderately well, but T5 performs noticeably better than the others, displaying perfect metrics for authentic news and high precision for fraudulent news. Figure 1 illustrates these trends, highlighting that most models perform reliably for true news but struggle with fake news, whereas T5 consistently outperforms the rest. This demonstrates the effectiveness of transformer-based prompting, particularly with T5, for Bengali fake news detection.

Fake News Detection:

In the category of traditional machine learning approaches, Logistic Regression produced the best outcome for fake news identification, reaching an accuracy of 75% and

Table 3: Performance comparison of traditional ML and Transformer-based models on Bengali fake news detection

Category	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Fake News	Logistic Regression	75	69	34	45
	Decision Tree	65	43	46	44
	Gradient Boosting	74	79	18	29
	Random Forest	73	65	24	35
	BERT	59	64	43	53
	RoBERTa	51	51	47	49
	T5	64	100	37	54
True News	Logistic Regression	75	76	84	45
	Decision Tree	65	75	73	74
	Gradient Boosting	74	73	98	84
	Random Forest	73	74	94	83
	BERT	59	57	76	65
	RoBERTa	51	51	55	53
	T5	64	100	100	100

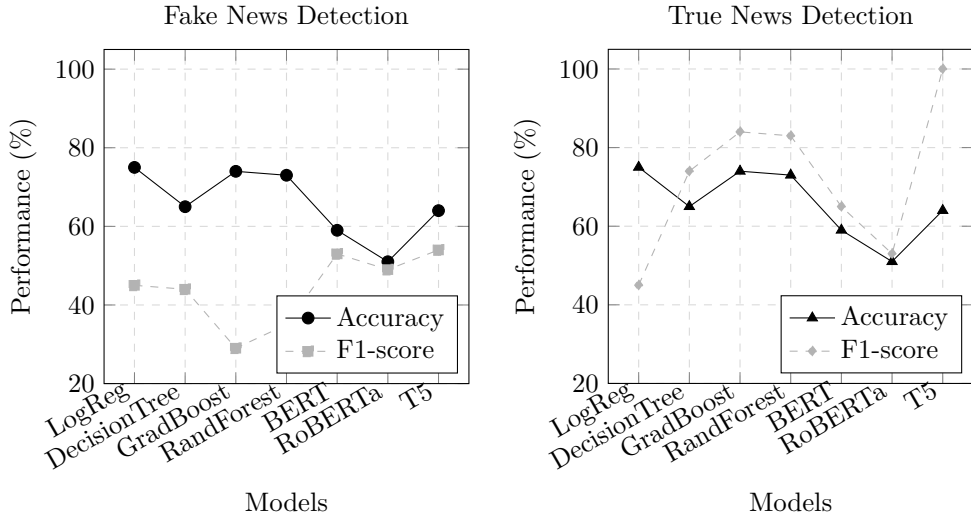


Figure 1: Performance comparison of traditional ML and transformer-based models on Bengali fake news detection. Left: fake-news task. Right: true-news task.

an F1-score of 45%. In contrast, the Decision Tree classifier was the weakest performer, with consistently lower scores across all evaluation metrics. Ensemble-based methods such as Gradient Boosting and Random Forest achieved intermediate results, with accuracy values between 73--74% and comparatively low F1-scores of 29% and 35%, respectively.

The transformer-based architectures displayed more diverse behavior. BERT obtained an accuracy of 59% with an F1-score of 53%. RoBERTa showed a slight advantage in recall (47%) but suffered from reduced precision (51%) and an F1-score of 49%, placing it below BERT overall.

The T5 model, however, stood out with notable improvements. It reached a precision of 100% and an F1-score of 54%. Although its recall was only 37%, the results emphasize T5’s effectiveness in correctly identifying fake news with almost no false positives, even if some cases were missed.

True News Detection:

The effectiveness of conventional machine learning in recognizing genuine news articles was demonstrated by Logistic Regression, which achieved 75% accuracy and a 45% F1 score. Ensemble techniques like as Random Forest and Gradient Boosting, on the other hand, produced better F1 scores of 84% and 83% as well as higher recall rates of 98% and 94%, respectively. Although it did not perform as well as the ensemble classifiers, the Decision Tree nevertheless did well. With 100% precision, recall, and F1 scores, T5 outperformed RoBERTa and BERT in transformer-based topologies, while their respective F1 scores were 53% and 65%.

All things considered, the results indicate that transformer-based systems specifically, T5 provide a more accurate and dependable framework for Bengali fake news identification. This finding highlights the value of big language models in addressing low-resource languages’ linguistic difficulties, where conventional feature-driven models frequently falter. The efficiency of several prompting techniques Zero-Shot, Few-Shot, and Chain-of-Thought across two tasks identifying Fake News and True News is compared in Table 4. Four metrics are used to analyze the three models (BERT, RoBERTa, and T5): accuracy, precision, recall, and F1-score. The T5 model with Chain-of-Thought prompting achieves the highest scores (Accuracy: 80%, Precision: 100%, Recall: 75%, F1-score: 85.7%) in the detection of bogus news, outperforming the other models. While RoBERTa’s Few-Shot prompting performs the lowest, it also excels in True News detection (Accuracy: 92.5%, Precision: 100%, Recall: 85%, F1 score: 91.9%). These findings demonstrate how well Chain-of-Thought prompting works for Bengali news identification, especially when combined with the T5 model.

Table 4: Performance of Prompting Techniques on Bengali Fake News Detection

Category	Prompting Technique	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Fake News	Zero-Shot	BERT	70	74	62	67.5
	Few-Shot	RoBERTa	54.5	55	47	50.7
	Chain-of-Thought	T5	80	100	75	85.7
True News	Zero-Shot	BERT	60	64	43	53
	Few-Shot	RoBERTa	51	51	47	49
	Chain-of-Thought	T5	92.5	100	85	91.9

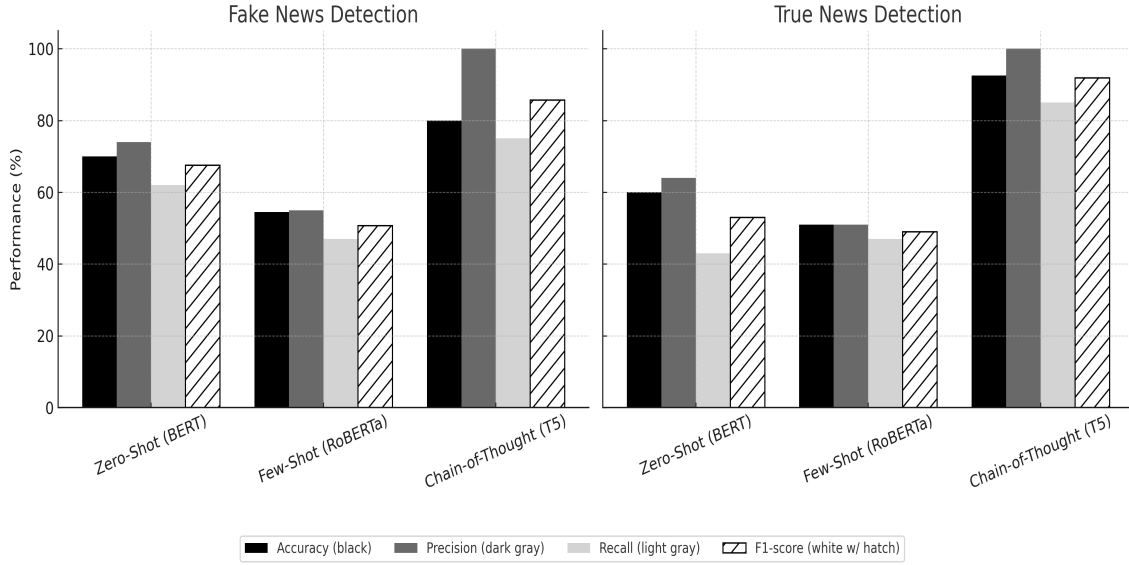


Figure 2: Performance of Prompting Techniques on Bengali Fake News Detection

The Figure 2 illustrates how different prompting techniques that is Zero-Shot, Few-Shot, and Chain-of-Thought that performs when applied to three transformer models: BERT, RoBERTa, and T5, for both Fake News and True News detection tasks.

A closer examination of misclassified examples revealed recurring patterns.

- Satirical or humorous content was frequently misclassified as true news, as the models struggled to distinguish exaggeration and irony from factual reporting.
- Politically charged headlines with ambiguous wording were often labeled incorrectly, particularly when lacking clear factual anchors.
- Code-mixed Bengali-English content created tokenization issues, leading to inconsistent predictions.

For instance, the headline “সরকার ঘোষণা করল আগামীকাল থেকে সব ঋণ মার্ফ” (“Government announces all loans waived from tomorrow”) was predicted as true by BERT due to its authoritative tone, despite being fabricated. Conversely, T5 with chain-of-thought prompting accurately judged it to be a fake based on its reasoning about it being implausible. Similarly, a satirical headline like “বৃষ্টি হলে রোদ পোহানো যাবে না, নতুন আইন পাশ” (“New law passed: No sunbathing allowed when it rains”) was misclassified by Random Forest, which relied on word-level patterns rather than contextual understanding.

These results indicate that prompt-based approaches, especially chain-of-thought prompting, yield stronger reasoning capacities, enhancing robustness and interpretability in detecting subtle misinformation.

8 Conclusion

This paper investigated the identification of fake news in Bengali through quick engineering, transformer architectures, and traditional machine learning architectures. The research experimented with a variety of techniques, from ensemble methods and logistic regression to advanced transformers such as BERT, RoBERTa, and T5, on a balanced dataset of Bengali news articles. It was revealed that traditional classifiers have poor memory and resistance to fake news, even though they are moderately accurate. Transformer models, particularly T5, however, showed significant performance improvements, especially when combined with chain-of-thought prompting. This showed that prompt engineering works even under low-resource conditions without the need for extensive fine-tuning. The experiments also showed several challenges, including inadequate recall for context-dependent or subtle disinformation and issues with processing satirical or code-mixed texts. These difficulties highlight the need for more linguistically diverse training data as well as reasoning-oriented prompting techniques. Future work will focus on improving prompt-based methods for cross-lingual and multilingual applications, introducing multimodal signals such as images and metadata, and expanding datasets with a variety of topical covers. The goal of the suggested framework is to offer a flexible and scalable approach to misinformation detection in Bengali and other languages with limited resources.

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